
Verification Belongs in the Laboratory: Scalable Oversight and Steering for Machine-Generated Science

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Abstract

Increasingly autonomous research systems that propose, execute and interpret their own experiments are being built across chemistry, materials and biology. Verification of these machine-generated science is usually attempted over published literature. We take it to belong in the laboratory, where those results are produced and the measurement that would settle a claim can be made. This paper defines a verification contract, which fixes what counts as having checked a claim, and builds a benchmark on a published dispute over an autonomous laboratory’s synthesis claims: 95 targets in six groups. It tests whether a verifier, given only the deposited evidence the experts held, reaches the findings they published. One architecture satisfies the contract: a model proposes checks and rival hypotheses, and every verdict is re-established by a small independent, trusted kernel. On the 20 stage-one targets the claim alone cannot reach, the layer reaches 20; the best unlayered condition at matched output reaches 14. No verifier, including the system the benchmark scores against, reproduces that system’s decisions from the evidence available when the claims were made. The code that must be trusted for a refutation to count grows with the variety of the record rather than its volume, and what the layer asks for next depends on the evidence it holds. The first is what scalable oversight requires; the second is what steering consumes. Both are properties of what a verdict costs in observations, adjudicative work and trusted code. This paper calls that cost alethic complexity and claims it is measurable. Here the trusted code is measured, the observations in part, and the adjudicative work not yet. Mathematics is coming to rely on proof assistants where the prover is a machine; science needs the same formalisation where the scientist is one. This is a first account of what it costs to turn what is true into what is known.

1 Introduction

Progress on materials for energy storage and carbon capture is gated by experimental throughput: how quickly a candidate can be made, measured and confirmed [Abolhasani and Kumacheva, 2023]. Across science, attention, experimental capacity and judgment are the scarce quantities, and their allocation is steered by institutions built for human-paced production: refereeing, funding, replication, editorial selection and taught taste [Simon, 1971, Kitcher, 1990, Weisberg and Muldoon, 2009]. Those institutions read prose, at publication cadence, and prose does not carry the objects an allocation decision consumes. When claims arrive faster than the institutions can read them, allocation passes by default to the generators themselves [Agrawal et al., 2024, Tao, 2026]. The question becomes which parts of the scientific landscape should be explored, challenged, replicated or left alone, and by what.

The generators are increasingly machines. Autonomous research systems that propose, execute and interpret their own experiments are being built across chemistry, materials and biology [Abolhasani and Kumacheva, 2023, Ghareeb et al., 2025, Mitchener et al., 2025, Ding et al., 2026]. They differ in where a person sits. Some write and run computational studies, where a result is a computation and checking it means re-running it [Lu et al., 2024]. Some propose hypotheses for people to test, so a person gates what is executed [Google, 2025]. Autonomous laboratories go furthest: the machine performs the measurement, interprets it and chooses the next one, and review arrives afterwards [Szymanski et al., 2023].

A result is established once someone other than its producer has checked it. Peer review is one such check, and it runs at publication cadence. Machine-generated science has a verification bottleneck [Cornelio et al., 2025], and the systems built to address it work over the record as published: they retrieve documents, test entailment, or re-run analyses on data that already exists [Thorne et al., 2018, Pan et al., 2023, Huang et al., 2025]. Inside the research systems themselves, the check is usually the generator's own. A survey of twenty-four such systems finds that seven of the nine closing an experimental loop decide what to try next on their own score, and that none built on a language model closes the loop on an external check [Ding et al., 2026].

Oversight scales only when checking a claim costs less than producing it, which is what makes a weaker overseer viable [Bowman et al., 2022]. For code and mathematics, checking is computing. For a claim about the world, the check needs the evidence the claim rests on, and much of that evidence never reaches the record. Publication filters it: the share of papers reporting positive results rose by more than twenty-two per cent between 1990 and 2007 [Fanelli, 2012, Journal of Negative Results, 2026]. What is filtered out carries signal, since a model trained on recovered notebooks of failed syntheses outperformed human intuition in choosing reaction conditions [Raccuglia et al., 2016]. Refutations, abstentions and the rivals a result had to rule out exist at the moment of adjudication and are rarely kept after it; a verifier has to be present then to keep them. Autonomous laboratories are therefore what makes verification urgent and what makes it possible: they produce results faster than those results can be judged, they are where that evidence still exists, and they are where the measurement that would settle a claim can be performed.

Where a claim is checked is one part of a larger question: what it costs to turn what is true into what is known. The cost has components: the observations a verdict needs, the adjudicative work that discharges it, and the code that must be trusted for the verdict to count. This paper calls it alethic complexity.¹ Two of its properties matter most for machine-generated science. Its growth decides whether oversight scales: whether the code that must be trusted grows with the number of claims or only with the kinds of evidence. Its allocation is steering: choosing the next experiment is a decision about where the cost is spent.

Systems are being built to make that steering decision by reading the published literature [Ghareeb et al., 2025, Mitchener et al., 2025]. A verification layer produces what that decision consumes: a verdict, the rivals that survived it, and the measurement that would separate them. It does not make the decision, in the way a proof assistant checks arguments without determining which theorems mathematics should pursue [de Moura and Ullrich, 2021]. Mathematics is coming to rely on proof assistants where the prover is a machine [Anthropic, 2026]; science needs the same formalisation where the scientist is one. The open problem is to design and evaluate verification-conditioned steering policies: policies that decide which experiments to perform, which claims to challenge or replicate, and which directions to defer, while keeping those decisions auditable.

Autonomous laboratories make this a safety problem as well. Their outputs become inputs to subsequent physical actions without waiting for publication or review, so an erroneous interpretation can determine what is synthesised next. A verification layer supplies what a safety policy needs and does not itself provide: a point at which a proposed action can be stopped, and the evidence trail on which stopping it can be justified [Zhang et al., 2026, Leong et al., 2025]. That matters most in areas such as biosecurity [Urbina et al., 2022]. The interfaces that make autonomous execution possible, including plans, instrument commands, logs and provenance, are what would make such a point enforceable [Moreau and Missier, 2013]; whether they suffice is open.

The instance measured here is a dispute over one autonomous laboratory's claims. In November 2023 that laboratory reported synthesising 41 new inorganic compounds in seventeen days [Szymanski

¹The adjective is used in its philosophical sense, concerning truth as against justification.

et al., 2023]. Within months an independent re-examination of the deposited evidence disputed much of the record [Leeman et al., 2024], and in January 2026 the original authors’ correction adjudicated 40 of the claims, confirming 36 and finding 4 inconclusive from the available X-ray analysis [Szymanski et al., 2026]. The disagreement ran compound by compound, between deposited evidence and named rival explanations. Claims, evidence, rivals and verdicts are the resolution at which this paper’s objects are defined, and both adjudicating sources published their reasoning at that granularity, which is what supplies per-claim reference labels. The deposits are public, so the record rebuilds and the grounding can be checked by the same criteria the paper applies to a verifier.

This paper argues where the verification of machine-generated science belongs, and demonstrates one architecture meeting a contract for it on one record. It contributes three things. First, a verification contract, which fixes what counts as having checked a claim, and a benchmark built on it, where an outside submission is scored by the same scorer as the runs reported here. The benchmark supplies 55 targets from the evidence deposited with the original claims and 40 more from an analysis performed after the dispute.

Second, one architecture, a verification layer that satisfies the contract, and what it measures of alethic complexity: the trusted-code component directly, and the observations component in part. A model proposes and a small independent component decides. With opus5, the layered architecture reaches all 20 informative stage-one targets (Section 4.2), where the best unlayered condition at matched output reaches 14. What it asks for next depends on the evidence it already holds. The component that must be trusted for a refutation to count grows with the variety of the record rather than its volume.

Third, a finding about what verification needs. Working from the evidence available when the claims were made, no verifier reproduces the pass/flag decisions of CARTOGRAPH [Shah and Kartik, 2026], the comparator stage two is scored against, including CARTOGRAPH itself. Granted the analysis the layer asked for, a re-refinement of the data already deposited, it agrees with the comparator on all 40 compounds. A laboratory is where such an analysis can be asked for while the claim is being made.

Those measurements are one record. Three conjectures generalise them, each a claim about alethic complexity: that its trusted component can be built at all for empirical claims, that it grows with the variety of a record and not its volume wherever the record comes from, and that verification inside the laboratory reaches parts of it that verification over the published literature cannot. Section 5 states each with what would refute it.

2 The benchmark

The benchmark asks whether a verifier, given only the deposited evidence an expert panel held, independently reaches the findings that panel published.

The A-lab episode is a claim-level dispute in which both sides published per-claim reasoning, so the record rebuilds end to end: the original synthesis claims [Szymanski et al., 2023], the published critique of them [Leeman et al., 2024], and the Correction that adjudicated each claim individually [Szymanski et al., 2026].

2.1 The verification contract

A claim is one assertion the source makes about one compound, together with the evidence deposited alongside it. A rival is an alternative account of that same evidence, which a verifier deciding for the claim would have to rule out. A verifier declares which rivals it considered; the contract does not fix how they are chosen.

A verifier has five verdict/outcome states: REFUTED, UNDECIDABLE-AT-BUDGET, BOUND-INCONCLUSIVE, CONSISTENT and SET-INADEQUATE. They don’t affirm a claim as a refutation cites evidence a third party can re-examine, whereas an affirmation doesn’t. This grammar inherits the falsifiability asymmetry [Popper, 1959].

REFUTED records that a check found evidence that contradict the claim. CONSISTENT records that nothing contradicted the claim, and that the measurement available was enough to separate it from the rivals. SET-INADEQUATE records that neither the claim or its rivals together explain the analysis, and do not account for what was measured. UNDECIDABLE-AT-BUDGET records that the measurement

available could not have separated the claim from the rivals. BOUND-INCONCLUSIVE records that neither of those last two could be shown.

Every verdict declares the conditions under which it holds: the rival class it was decided against, the instrument model assumed, the evidence stage held, the measurement budget, and the error rate. The evidence stage is the benchmark’s, since it controls what is released; the other four are the submission’s, and the budget is declared per instance rather than fixed across the population.

A refutation counts only when an independent checker re-establishes it [Thorne et al., 2018]. Validity is an established contradiction between the deposited evidence and the claim record. The requirement is read from the claim record rather than from the witness text, because that text is written by the system being judged. The checker uses the standard library only and shares no code with any verifier.

An outside verifier needs none of this project’s layer, models, primitives or code. It submits one JSONL file, and the same scorer and rubric that produced the results in Section 4 produce its score. The benchmark distribution gives the submission schema and the classes of evidence a submission reports on.

2.2 The targets

The conditions for a target is that an expert source published it, and a model-written check could reach it from the claim and the deposited evidence (Figure 1). The evaluation population is the 40 claims the Correction adjudicated. **K1–K5** are the critic’s results, and **K6** is an external comparator’s decision computed from the correction’s refinement data. A deterministic harness re-derives every target from the deposit alone. The exclusions, the populations, and the harness are in the benchmark distribution.

	SOURCE	EXPERT DERIVATION	TARGETS	STAGE
K1 structure mismatch	automated structure files	critique Table I, the E2 flag	7	one
K2 symmetry	automated structure files	critique Table III, indexed_sym	38	one
K3 novelty	reference databases	critique Table I, the E4 flag	2	one
K4 lattice parameter	automated structure files	critique Table II	5	one
K5 mixed sites	automated structure files	critique section IV prose	3	one
K6 baseline decision	manual refinement columns	40 decisions, 8 flagged	40	two
55 stage-one targets, 40 stage-two				

Figure 1: The ceiling. Each row reads left to right: the finding, its source in the record, the expert derivation, the number of targets surviving exclusion, and the evidence stage.

Evidence stages Stage one is the evidence that existed when the claims were made. Stage two is an expert re-refinement that exists because the dispute happened, and it supplies a re-refinement against an expanded candidate phase set. A verifier that abstains and does not reach CONSISTENT, as well as also requesting the analysis type stage two holds receives it. Note that stage-two evidence is contaminated, as it has been produced by people who knew the outcome, and scored separately.

K6 is the only group scored at stage two. It is the pass/flag decision of CARTOGRAPH [Shah and Kartik, 2026], reproduced from that paper and computed from the correction’s manual refinement columns, so it could not have been computed at stage one. Agreement on K6 is agreement with that decision, which is a different label from the correction’s confirmed and inconclusive statuses.

2.3 Scoring

Scoring is per target. Five outcomes are recorded. RECOVERED: the target was reached and matched. MISSED: a check reached it and gave a different answer. UNVALIDATED: a refutation was made and

the scorer could not read it. NOT-OBSERVED: a check ran and recorded nothing for that quantity. NOT-ATTEMPTED: no check that could reach the target was written.

2.4 Controls

No expert label, flag, verdict or finding appears in any input, and each compound is submitted once.

Contamination. Two directions are constructed from the evaluation set with the compound name preserved in both. H1 removes a fault from a compound the critique flagged; B1 injects one into a compound nobody flagged. The directions predict opposite behaviour, which is what makes this a control for memorisation: a verifier that works answers H1 `cannot_verify` and B1 `refuted`, while one recalling the literature does the reverse.

Mutants. A second control set is wrong by construction, with the truth recorded outside the claim by a committed generator, across six families spanning four orders of magnitude of difficulty. Detecting a planted fault and reproducing an expert finding are different questions, and the mutants are not reproduction targets.

3 Aletheia: the verification layer

Aletheia is the instance of the architecture measured here: one primitive library, one admission library, one forward-model library, one pricing library and one checker, filled with what a single crystallographic dispute required. The architecture is the separation of proposing from deciding. Aletheia is one filling of it. The results in Section 4 are Aletheia’s; the conjectures in Section 5 are the architecture’s.

3.1 Design

The layer is built on the asymmetry of scalable oversight, between a system and the weaker overseer that checks it [Bowman et al., 2022]: an overseer can be weaker than what it oversees when checking is cheaper than producing. So the layer separates proposing from deciding. A model writes checks and proposes rivals, and it is untrusted until a program sharing no code with it re-establishes the result.

Proof assistants have a similar shape: a kernel small enough to audit, proof terms that are re-checkable objects, and search procedures that may be arbitrary because nothing they produce is trusted until the kernel accepts it [Barendregt and Geuvers, 2001]. Certifying algorithms and proof-carrying code are the same move outside logic [McConnell et al., 2011, Nacula, 1997].

A proof kernel in maths cannot be fooled as mathematics only needs to be internally consistent. An empirical kernel executes under assumptions it cannot verify. So constructing trusted kernels for empirical science is an open problem. Such a kernel would have to reach the experiment, where the only trusted kernel is nature itself. Since the benchmark is retrospective, it is not done here. Autonomous laboratories are what would put an experiment within a verifier’s reach, it is both the motivation for needing scalable oversight, as well as the only place that can substantiate it. Section 5 states the problem.

Rivals are designed from the framework of multiple working hypotheses, where a claim is supported to the extent that alternative hypotheses have been excluded [Chamberlin, 1890, Platt, 1964].

Abstention is the default output, since unfalsified is the default condition of a claim [Chow, 1970, El-Yaniv and Wiener, 2010], and an abstention carries the request for evidence on which the second evidence stage is conditioned.

3.2 The chain

The layer takes one claim, and returns a vector, a verdict, the conditions it holds under, and the residual, which is the rivals still standing together with the observation that would separate them.

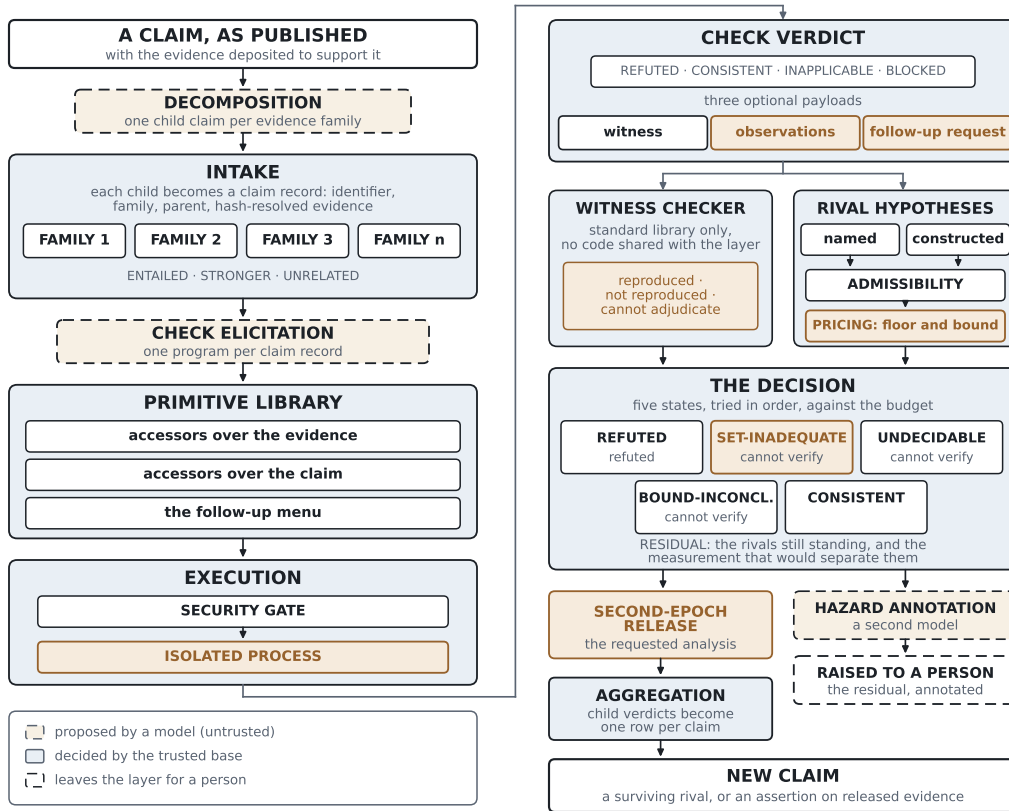


Figure 2: A demonstration of what the layer should consist of. The entire chain, from claim to claim. Each step is described in Section ?? . Hazard annotation does not yet exist, and is more relevant in biosecurity venues than material science ones. Read down each column and then across: vertical position is sequence and horizontal position is alternatives.

Decomposition and intake. A model decomposes a single claim into several child claims, each asserting one thing about one family of evidence (here as the phase present, the cation ordering, the weight fractions, or novelty against the reference databases). This is done such that an assertion is decomposed into units checked independently from established text [Wei et al., 2024]; here the library reads a physical deposit. Each unit becomes a record carrying an identifier, a pointer to its parent, and the evidence it rests on, resolved to specific deposited artifacts and hashed [Kuhn and Dumontier, 2014]. Recording which artifacts a claim rests on, so that a conclusion can be traced back to them, is standardised as a provenance model [Moreau and Missier, 2013] and implemented for autonomous research as a chain of evidence [Meng et al., 2026]. How finely a claim splits varies several-fold across models, and Section 4 reports that spread.

Elicitation. A model writes one program per child claim. Each program runs once. Agents already generate re-runnable hypothesis tests [Miao et al., 2026] and design sequential falsification experiments under formal error control [Huang et al., 2025]. This layer seals the elicitation, where a program reads the deposit only through a fixed set of functions, the primitive library of Figure 2, which bounds what any check can look at and returns measurements. Programs are stored and reused, so every later stage can be changed and re-run without another model call; a program is rewritten only when the set of functions changes.

Execution. Anything outside the primitive library is refused before the program runs. Across 808 executed checks the gate refused sixteen programs, each calling a function that does not exist, and the refused names are a record of what the vocabulary lacks and a reviewer can read them as candidates for the next version. The library has grown this way: three functions added at one point made three

target groups reachable that no check could previously ask about. A program then separately runs the check, without credentials and under a time limit.

Check Verdict. A check returns one of three statuses: refuted, consistent and inapplicable, with a payload. A refutation carries a witness, which records where the check looked, what it found there, and what the claim required instead; that is the object the independent checker re-establishes. Any check may also record observations, which are measured values with no refutations attached. This is the common element across operational evidence graphs [Ren et al., 2026, Meng et al., 2026].

A check may also name the follow-up measurement that would settle the question, drawn from a fixed menu. Choosing the next observation to discriminate hypotheses is owned by active and sequential testing and by Bayesian experimental design [Chernoff, 1959, Naghshvar and Javidi, 2013, Chaloner and Verdinelli, 1995], and previous work in steering, and stopping in an autonomous laboratory includes [Shah and Kartik, 2026, Kusne and McDannald, 2026].

Rival hypotheses and AI Scientists. A rival is an alternative account of the same evidence. The record names some, as provided by the original papers here; where it names none, the layer constructs one by disordering the claim’s own structure.

More importantly, the slot is also where a new hypothesis enters. Any hypothesis worth proposing contradicts something, whether an account the record holds or the absence of one, so it enters as an alternative to what the record currently says. That makes the rival slot an interface for hypothesis generation, which is where autonomous-science systems concentrate their effort [Ghareeb et al., 2025, Mitchener et al., 2025, Lu et al., 2024, Google, 2025]: a proposal may come from any system, is admitted and priced like any other, and is excluded from the verdict. High-throughput AI generated science can therefore be used without being trusted. Generating those alternatives from the data is an active research area, and automated multiple-hypothesis phase identification is the nearest attempt [Fei et al., 2025] from the group whose claims this benchmark rebuilds [Szymanski et al., 2023]; the layer accepts its output as a rival source and has run it.

Admission and pricing. A rival is admitted when it passes admissibility tests, the trusted base’s equivalent of the primitive library. The tests check form rather than scientific plausibility, so a proposer can inflate a claim’s apparent difficulty with a well-formed implausible rival. Each admitted rival is then priced against the claim. Two quantities travel with the pair: the achievable bound says a measurement of a given size is enough to separate them, and the floor says none below it can [Chernoff, 1952, Le Cam, 1973, Tsybakov, 2009]. For simulated profiles p and q with squared Hellinger distance $H^2 = 1 - \sum_k \sqrt{p_k q_k}$, error rate δ and m rivals, the achievable bound is $\ln(m/\delta)/H^2$ detected counts and the floor is $\ln(1/(4\delta(1 - \delta)))/(2 \ln(1/(1 - H^2)))$. Both hold for two fixed structures under counting noise alone, with the instrument assumed. Choosing an experiment by how much it would discriminate is established [Chaloner and Verdinelli, 1995], and relating how much measurement is taken to what it can resolve is long-standing in diffraction [Hegerl and Hoppe, 1976].

The instance decision. Each child claim receives one of the five states of Section 2.1 by an ordered rule, first match wins: a witnessed refutation, then set-inadequacy, then the three pricing regions, which are the floor above what the measurement provides, the achievable bound within it, and neither. Set-inadequacy is asked before the pricing regions because if the offered hypotheses do not account for the sample, pricing claim against rival compares two explanations neither of which explains what was measured. This state is computed by the trusted base.

Aggregation. Aggregation collects the verdicts on every claim and every compound into one place. Each verdict carries a residual stating what remains open on that claim and what would settle it, and those are collected with them. A choice of next experiment reads both. The layer currently writes this as one row per claim.

How the space should be represented is open. Typed evidence graphs are one attempt [Ren et al., 2026, Meng et al., 2026], and the claim-graph lineage of nanopublications, micropublications and provenance vocabularies supplies the substrate without a kernel that decides [Groth et al., 2010, Kuhn and Dumontier, 2014, Clark et al., 2014, Moreau and Missier, 2013]. How to act on it is a second open problem; retrieval-augmented generation [Lewis et al., 2020] and reinforcement learning with

verifiable rewards [Shao et al., 2024, Lambert et al., 2024] are the obvious first attempts. Section 5 returns to both.

Residual. The residual leaves the layer naming a measurement or experiment that the lab should perform next. Whether that measurement is hazardous is not a judgement this layer currently makes: hazard is a domain judgement, and the trusted base holds only what is small enough to read. The layer annotates the residual and does not gate it. What stops a proposal is a separate authority acting on the record the layer produces, and the composition of the two is open (Section 5).

3.3 The trusted base

Five of the layer’s components are libraries. The primitive library holds the functions a check may call to read evidence. The admission library holds the tests that decide whether a proposed rival hypothesis is valid. The forward-model library holds the methods that turn a hypothesis into a predicted observation. The pricing library holds the methods that say how much observation separates two hypotheses. The checker holds one predicate per class of evidence. The decision rule and the aggregation rule are fixed. Aletheia’s trusted base is 3,757 lines.

A record drawing on other instruments or other kinds of claim adds entries to these libraries. This is by design, these libraries should continue to be expanded, but expanded in versions, as well as upheld openly, sourced communally, and remain human-legible. How trustable these libraries are, as well as how they scale across domains remain as open problems, see Section 5.

4 Results

4.1 Conditions

Four models ran through Aletheia on the same 40 compounds using stage-one evidence. Three unlayered conditions ran on one of those models: reading the deposited file with no execution; writing and running one program without seeing the file; and writing and running one program with the file supplied. Six mutant compounds ran through all seven configurations as a contamination control.

The file-seen condition was also run with seven independent attempts per compound, to achieve matched compute with the layer. Its attempts are scored by the layer’s rule: a flag counts if any attempt’s refutation is re-established by the checker, and a value counts if any attempt records a correct conversion of the deposit.

A second round ran on one model with stage-two evidence readable by eliciting follow-up checks. It is a different population from the four-model comparison.

4.2 Target recovery

Of the 55 stage-one targets, 35 are K2’s echo half, where the deposited symmetry equals the claimed one, so reporting the claim earns them; every condition that sees the file scores between 31 and 35 there. We call the remaining 20 informative: the claim alone does not reach them, and reaching them requires the deposit or the reference databases.

The layered opus5 run reaches all 20 informative targets (Table 1). The file-seen condition reaches 6 in a single attempt. Given seven attempts at matched output it reaches 14, and none of its attempts reaches K3. Within that run the first attempt alone reaches 10, so the six further attempts add four targets, three of them on K1.

Across the four models, only K1 and K4 differ, at 7/5/2/3 and 5/0/1/1; K2, K3 and K5 return identical scores for every model, each at its ceiling. Only opus5 reaches the ceiling through the layer; the other three reach 11 to 13 informative targets.

On the echo half, the blind condition scores 20, below both reading without a program (31) and a program that sees the file (34): writing a program against a file it cannot inspect costs more than not writing one at all.

Table 1: Target recovery. The two blocks share the group spine, and the layered cell appears in both. Group denominators are not pooled, except in the informative subtotal, which sums the five groups outside K2’s echo half and is the figure the abstract cites. The matched column is the file-seen condition given seven attempts per compound at 0.95 times the layer’s output tokens, scored by the same any-attempt rule as the layer (Section 4.1). Bold marks a cell that exceeds every other cell in its row within the block. K6 is scored at stage two and is reported in Table 4; it has not been elicited for the other three models. The one target the single-attempt file-seen condition did not attempt is counted against the benchmark’s denominator. Every cell is one model and one run, with no intervals.

Block A — four models, all through the layer

group	targets	opus5	sonnet5	gpt56	gpt56-terra
K1 file differs	7	7	5	2	3
K2 determination = claim	35	35	35	35	35
K2 requires the file	3	3	3	3	3
K3 already known	2	2	2	2	2
K4 lattice parameter	5	5	0	1	1
K5 prose	3	3	3	3	3
informative subtotal	20	20	13	11	12

Block B — one model (opus5), five conditions

group	targets	reading only	one program file unseen	one program file seen	file seen ×7, matched	the layer
K1 file differs	7	4	5	2	6	7
K2 determination = claim	35	31	20	34	35	35
K2 requires the file	3	2	3	2	3	3
K3 already known	2	0	0	0	0	2
K4 lattice parameter	5	1	0	0	2	5
K5 prose	3	3	3	2	3	3
informative subtotal	20	10	11	6	14	20

The reported single attempt and the first attempt of the seven are the same condition elicited twice, and they differ by four informative targets. A single-attempt column should be read with that spread in mind.

4.3 Contamination control

Six of the seven configurations are correct on both healed compounds; the one miss is the condition that never saw the file, which refuted on the deposited P 1 header (Table 2). The remaining failures fall in the detection direction, where injected disorder on three of four compounds goes unreported by two models.

Table 2: Contamination control. Seven configurations over six constructed compounds; completion and accuracy are reported separately. H1 is the memorisation-sensitive direction. B1 is coarser than the benchmark, at eight to twelve shared cation sites, so a high B1 score does not indicate that the benchmark is easy.

configuration	checks completed	H1 correct	B1 correct
reading only	6 of 6	2 of 2	4 of 4
one program, file unseen	6 of 6	1 of 2	4 of 4
one program, file seen	6 of 6	2 of 2	4 of 4
layer, opus5	6 of 6	2 of 2	4 of 4
layer, sonnet5	6 of 6	2 of 2	2 of 4
layer, gpt56	6 of 6	2 of 2	1 of 4
layer, gpt56-terra	6 of 6	2 of 2	1 of 4

Table 3: What follow-up request the layer asks for, by what it already holds. Columns differ in whether the re-refinement had already been released to that child.

modality requested	lacks it	holds it
re-refinement, expanded phase set	75	11
higher-resolution XRD	3	54
elemental analysis	1	14
electron diffraction	0	4
neutron powder	0	3
longer counting	0	1
total	79	87

Table 4: The second evidence stage. Same claims, same model, same population; the two rounds differ in whether the requested analysis was available when the checks were written.

	round one	round two
checks written	blind to stage two	with stage two granted
agreed where the comparator flags	4 of 7	8 of 8
agreed where it does not flag	27 of 31	32 of 32
overall	31 of 40	40 of 40
recorded no usable observation	2	0

4.4 Layer output

Over 808 child instances pooled across the four models, four of the five instance states occur: BOUND-INCONCLUSIVE 382, CONSISTENT 203, REFUTED 185 and SET-INADEQUATE 38. UNDECIDABLE-AT-BUDGET is the only state the floor certifies, and it occurs zero times. The floor carries a calibration constant, and that constant has not been calibrated against this instrument. Because the benchmark is retrospective, no measurement can be commissioned to calibrate it, so the constant is set where it cannot fire spuriously; the price of that setting is that it does not fire at all. Calibrating a floor against a named instrument is one of the open problems of Section 5.

Pricing runs on 214 of the 808, on all 203 CONSISTENT instances and on 11 of the 38 SET-INADEQUATE, approximately a quarter of total.

Four of the forty claims carry a rival that a person proposed; for the rest the layer constructs an alternative by disordering the claim’s own structure. The tool-proposed path was exercised, with an external phase-identification system: its proposals were admitted, priced and reported.

4.5 Steering

When the layer chooses an abstaining verdict, it names the follow up request that would settle the question, committed in the same completion as the check. What it names depends on what it already has. Children that had not been given the re-refinement asked for it in 75 of 79 cases; children that had been given it asked for it in 11 of 87, and asked instead for higher-resolution diffraction in 54 (Table 3).

To simulate steering, stage-two evidence was hidden and released only on request.

K6 is scored on stage-two evidence. The layer worked on stage-one evidence, abstained, named the analysis that would settle the question, was granted it, and was scored against the external comparator’s decision. Round one scores 31 of 40 and round two 40 of 40, agreeing on all 8 compounds the comparator flags and all 32 it does not (Table 4).

4.6 Scalable Oversight

In Aletheia, what must be trusted for a refutation to count is 1,015 lines using the standard library only, sharing no code with the layer or with the models that wrote the checks. It grows with the number of evidence classes rather than the number of claims: one predicate per class, and adding

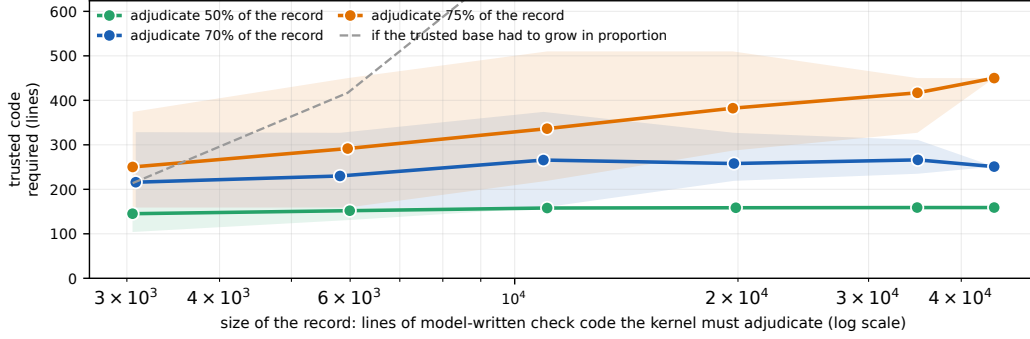


Figure 3: The trusted base against the record it adjudicates. For each record size, the smallest set of checker predicates that adjudicates the stated share, averaged over repeated subsets of the record, with the shaded band covering the tenth to ninetieth percentile. The dashed line is proportional growth.

compounds adds none. Figure 3 shows the consequence: the trusted code needed to adjudicate a fixed share of the record stays flat as the record grows.

Once compiled, a check is stored, so changing the decision rule and re-adjudicating the whole record costs no model calls.

4.7 Limitations

The contract specifies five verdict states; this record exercises four. Pricing runs on 214 of 808 instances and the floor certifies none of them, so the measurement-budget half of the contract is demonstrated as a mechanism and not evaluated as a method. What the layer is shown to do here is decompose, seal, re-establish, and abstain with a request. What it is not shown to do is price a claim against a rival well enough for the price to decide anything.

The evidence is one episode and one measurement class, with limited data. So several of the comparisons above rest on populations too small to reach statistical significance, and are reported as consistent with a reading rather than as establishing one.

5 Outlook and conclusions

Alethic complexity. Most of the problems below are questions about alethic complexity, the cost of turning what is true into what is known. Hypothesis generation prices its observations. Representation and steering organise and allocate its adjudicative work. The libraries and the kernel are its trusted code, and scalability is how that code grows. This paper measures the trusted code, part of the observations, and none of the adjudicative work. A measure of the whole needs all three on one scale for a single claim.

Hypothesis generation. The proposer sits outside the trusted base and what it proposes is priced. Pricing gives a proposer an objective, since for any admissible rival the trusted base returns the measurement that would separate it from the claim, and proposing then becomes a search for the admissible alternative that costs the most to exclude. That objective is a number, so a proposer can be trained against it. The open problem is an admission gate that holds against models that game it [Anil et al., 2021, Skalse et al., 2022].

Representation. Aggregating into one row per claim cannot express that a verdict on one claim obliges a re-check of another. Which verdicts carry that obligation and how far it propagates is what a representation has to make computable, and it is the open question the typed-graph programmes are closest to answering [Ma, 2026].

Steering. Several claims stand open at once, each residual names a measurement, and a laboratory has one instrument. Choosing among residuals is an allocation problem this paper leaves untouched,

and the measurement budget should follow what a claim is worth rather than a fixed count per instance. The record a steering policy would train on is also different in kind from the published literature, because it keeps what publication filters out: refutations, abstentions, and the rivals that survived. Failed syntheses have already outperformed human intuition as training data once [Raccuglia et al., 2016]; whether a policy trained on an audited record steers better than one trained on the literature is a measurable question.

The libraries. This instance holds a primitive library, an admission library, a forward-model library, a pricing library and a checker, for one class of measurement. A record drawing on other instruments or other kinds of claim is adjudicated by adding entries, so the libraries are where the architecture grows, and how they grow determines what the architecture is worth. They should be open source, versioned, and kept by the communities whose measurements they encode, the way mathematics keeps a shared library of formal proofs [The mathlib Community, 2020]: a verdict cites the library version it was decided under, a library change names the verdicts it reopens, and what the trusted base contains is public in the same sense the claims it judges are. Each discipline already holds candidate entries. Crystallography has a long-established answer for comparing models that differ in the number of free parameters [Hamilton, 1965], and a pricing library should reach for it where it applies. What is open is transfer: whether libraries of this shape carry from one domain to another, and what a pricing library holding several statistical methods selects among them on.

Safety. An autonomous laboratory acts on its own results without waiting for review, so an interpretation determines what is synthesised next. The layer supplies the point at which a proposed action can be stopped and the evidence trail on which stopping it can be justified. Hazard models, containment and instrument state sit outside it [Zhang et al., 2026, Leong et al., 2025, Tang et al., 2025]. How an evidence-admission contract composes with an action-permission contract is open, and that composition decides whether an audited record can govern what a laboratory does next.

The kernel. Every artifact the layer reads stands in for a measurement made under conditions the layer cannot check. Calibration, specimen identity, instrument state and specimen history enter as declared assumptions. A kernel for empirical claims would have to reach the experiment, and an autonomous laboratory is what would put an experiment within a verifier’s reach. That is the first conjecture, that such a kernel can be built at all. An argument that no empirical primitive can authenticate itself would refute it.

Scalability. The checker holds one predicate per class of evidence, so the trusted code needed to adjudicate a fixed share of the record grows far more slowly than the record (Section 4.6). The second conjecture is that this holds wherever the record comes from: the cost of oversight tracks the variety of a record and not its volume. A domain in which the number of kinds of evidence grows with the number of claims would refute it.

The laboratory advantage. The third conjecture is that verification inside the laboratory reaches what verification over the published record cannot. The reason is what the record keeps. Refutations, abstentions and the rivals a result had to rule out exist when a claim is adjudicated and are rarely published, so a verifier over the record cannot see them and a verifier in the laboratory can keep them. The measurement behind it: at stage one no verifier reproduces the comparator’s decisions, and granted the analysis it asked for, a re-refinement of data already deposited, the layer agrees with the comparator on all forty compounds (Table 4). It has a formal ancestor, since for causal graphs verifying a proposed answer is proven to need fewer interventions than finding one [Choo et al., 2022], and interventions are the resource an autonomous laboratory spends. If the conjecture holds, verifiers belong where the discriminating measurement can be run, and a verification layer becomes standard laboratory equipment rather than a stage of publication. Two findings would refute it: a domain in which the discriminating experiment costs more than the claim is worth, and a domain in which the published record suffices at scale.

This paper defines a verification contract, builds a benchmark on an adjudicated dispute over an autonomous laboratory’s synthesis claims, and measures one architecture against it. The layered verifier reaches all 20 informative stage-one targets, where the best unlayered condition at matched output reaches 14. What it asks for next depends on the evidence it already holds, and the code that must be trusted for a refutation to count stays small as the record grows. A layer of this kind, running

inside the laboratory at the time of deposit, would have surfaced what a separate critique was later required to establish, and would have declined the questions that needed an experiment nobody had yet performed.

What is proposed here is an architecture rather than any one system. A model proposes, a small independent component decides, and everything that component must trust is a library indexed by kind and small enough to read. This specific benchmark fills those libraries with what one crystallographic dispute required. Aletheia is one system; what the architecture is worth depends on what fills the libraries next.

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The deposited evidence this benchmark rebuilds was made public by the authors of the original study and its critique, and the external comparator’s decisions are reproduced from its published definition.

The benchmark, scorer, and harness are at <https://github.com/ArdenTSH/verification-layer-benchmark>. The layer, libraries, and kernels are at https://github.com/ArdenTSH/Aletheia_v0

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